

	M.A.M. SCHOOL OF ENGINEERING		
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	Siruganur, Trichy 621 105	www.mamse.in	

Department of Electrical and Electronics Engineering
Academic Year 2025-2026 (Even Semester)

Unit - I

Sub.Code/Sub.Name : EE3033 Hybrid Energy Technology **Date** :
Year/Sem. : III/VI

	Part A (10 X 2 = 20 Marks) Answer All the Questions	M.A	CO	BTL
1.	List various renewable energy resources.	2	CO1	BTL1
2.	Discriminate spring and neap tides?	2	CO1	BTL1
3.	State the principle of power generation using tides?	2	CO1	BTL1
4.	What is greenhouse effect?	2	CO1	BTL1
5.	What is mean by global warming?	2	CO1	BTL1
6.	What are the advantages of Renewable Energy source?	2	CO1	BTL1
7.	What is meant by non –commercial energy?	2	CO1	BTL1
8.	List the non-conventional energy sources.	2	CO1	BTL1
9.	What are the factors causing winds?	2	CO1	BTL1
10.	Give any two environmental aspect of electric energy conversion.	2	CO1	BTL1

	Part B (5X16=80)	M.A	CO	BTL
11.	Explain in detail the Classification of Hybrid Energy Systems	16	CO1	BTL2
12.	Describe the principle of generation of Bio gas and mention the factors affecting its generation.	16	CO1	BTL2
13.	What are the types of ocean thermal energy conversion power plants? Describe in detail the Anderson OTEC cycle	16	CO1	BTL2
14.	Explain the design and principle of operation of general Fuel cell and Fossil Fuel cell.	16	CO1	BTL2
15.	i)Explain in details about the Geothermal Energy sources. ii)What is Hydrogen energy? Explain the operation of Hydrogen energy system with a neat schematic.	8 +8	CO1	BTL2

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Department of Electrical and Electronics Engineering
Academic Year 2025-2026(EVEN Semester)

Unit - II

Sub.Code/Sub.Name : EE3033 Hybrid Energy Technology :
Year/Sem. : III/VI :

	Part A (10 X 2 = 20 Marks) Answer All the Questions	M.A	CO	BTL
1.	Explain the principle of operation of induction generators?	2	CO2	BTL1
2.	Name any four types of generators used in wind energy conversion systems.	2	CO2	BTL4
3.	Show the merits of DFIG over SCIG for wind energy conversion?	2	CO2	BTL1
4.	Why are induction generators preferred over dc generators in WECS.	2	CO2	BTL1
5.	Compare the PMSG and IG used in WECS?	2	CO2	BTL4

	Part B	M.A	CO	BTL
11.	Draw the schematic of Permanent Magnet Synchronous generator and explain the construction and principle of operation in detail. Also discuss the characteristics and issues briefly	16	CO2	BTL2
12.	Explain the principle of operation and constructional features of squirrel cage induction generator with a neat diagram. Analyse the merits and demerits of the above.	16	CO2	BTL2
13.	Explain the analysis of Induction Generator used for Wind Energy Conversion System.	16	CO2	BTL2

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**Department of Electrical and Electronics Engineering
Academic Year 2025-2026(EVEN Semester)**

Unit - II

Sub.Code/Sub.Name : EE3033 Hybrid Energy Technology :
Year/Sem. : III/VI :

	Part A (10 X 2 = 20 Marks) Answer All the Questions	M.A	CO	BTL
1.	What is the principle of DFIG?	2	CO4	BTL1
2.	What are the merits of squirrel cage induction generator for wind energy conversion?	2	CO4	BTL1
3.	Differentiate between SCIG and DFIG.	2	CO4	BTL4
4.	Name any four types of generators used in wind energy conversion systems.	2	CO4	BTL1
5.	Why are induction generators preferred over DC generators in WECS?	2	CO4	BTL1

	Part B	M.A	CO	BTL
6.	Explain the analysis of Induction Generator used for Wind Energy Conversion System.	16	CO2	BTL2
7	Explain about DFIG based energy conversion system	16	CO2	BTL2

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Department of Electrical and Electronics Engineering
Academic Year 2025-2026(EVEN Semester)

Unit - III

Sub.Code/Sub.Name : EE3033 Hybrid Energy Technology
Year/Sem. : III/VI

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	Part A (5 X 2 = 10 Marks) Answer All the Questions	M.A	CO	BTL
1.	Draw and label the block diagram of solar photovoltaic system.	2	CO3	BTL1
2.	Discuss the use of line commutated converters in PV system. Draw the schematic diagram.	2	CO3	BTL1
3.	Generalize the significance of buck boost converter?	2	CO3	BTL3
4.	Give the schematic diagram of buck boost converter.	2	CO3	BTL1
5.	What is battery sizing? (or) Write about the aspects of battery sizing.	2	CO3	BTL1
6.	Show the weakness involved in sizing the solar arrays.	2	CO3	BTL1
7.	Identify the factors to be considered for the selection of inverter and batteries for solar energy conversion.	2	CO3	BTL1
8.	Define the photo conversion efficiency of the PV cell.	2	CO3	BTL1
9.	Define Depth of Discharge (DOD)	2	CO3	BTL1
10.	List the major components of solar PV system	2	CO3	BTL1

	Part B (5X16=80)	M.A	CO	BTL
11.	Explain the converters used for solar energy conversion.	16	CO3	BTL2
12.	Describe how a three-phase line commutated converter is operated as an inverter	16	CO3	BTL2
13.	Draw the block diagram of the solar PV system and explain the principle of operation in detail.	16	CO3	BTL2
14.	Explain in detail with block diagram i) OFF grid system ii) Grid tied system	8+8	CO3	BTL2
15.	Draw the schematic diagram of standalone solar photo voltaic system. Explain their functions.	8+8	CO3	BTL2

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Department of Electrical and Electronics Engineering
 Academic Year 2025-2026 (EVEN Semester)
 CIA – II Exam

Unit - IV

Sub.Code/Sub.Name : EE3023 Hybrid Energy Technology :
 Year/Sem. : III/VI :
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	Part A (10 X 2 = 20 Marks) Answer All the Questions	M.A	CO	BTL
1.	What is meant by matrix converters?	2	CO4	BTL1
2.	What are the applications of AC voltage controller or ac chopper	2	CO4	BTL1
3.	Where is matrix converters used	2	CO4	BTL1
4.	List the different types of PWM Inverters	2	CO4	BTL1
5.	Define Bidirectional converter.	2	CO4	BTL1
6.	What is the use of Inverter in Grid Connected PV system.	2	CO4	BTL1
7.	What Factors to be considered for selection of Grid Interactive inverters	2	CO4	BTL1
8.	Define grid integrated solar system.	2	CO4	BTL1
9.	List the applications of Grid Interactive Inverter PV system?	2	CO4	BTL1
10.	What will happen if no load is connected to a solar PV system?	2	CO4	BTL1

		Part B	M.A	CO	BTL
11.	a)	Write short notes on the grid interactive inverters	8+8	CO4	BTL 3
12.	a)	Explain the following in detail about AC voltage controller ii.	8+8	CO4	BTL 3
13.	a)	Explain in detail about Bidirectional DC-DC converter.with neat circuit diagram	16	CO4	BTL 3
14.	a)	Describe principle of operation of PWM inverter and describe how it is used for PV system	16	CO4	BTL 4
15.	a)	Explain the working and operation of Matrix Converter.	16	CO4	BTL

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Department of Electrical and Electronics Engineering
Academic Year 2025-2026 (EVEN Semester)

Unit - V

Sub.Code/Sub.Name : EE3023 Hybrid Energy Technology :
Year/Sem. : III/VI :

	Part A (10 X 2 = 20 Marks) Answer All the Questions	M.A	CO	BTL
1.	Define hybrid systems?	2	CO4	BTL1
2.	What is the need for hybrid energy systems.	2	CO4	BTL1
3.	List out some of the hybrid systems used in industries.	2	CO4	BTL1
4.	What are the disadvantages of Hybrid System?	2	CO4	BTL1
5.	What are the types of Hybrid System?	2	CO4	BTL1
6.	List the disadvantages of switched hybrid PV diesel system?	2	CO4	BTL1
7.	List the advantages of Parallel hybrid PV diesel system?	2	CO4	BTL1
8.	What are the major components of Biogas Solar thermal Hybrid System.	2	CO4	BTL1
9.	What is the use of Solar dryer in Solar Biomass Hybrid System?	2	CO4	BTL1
10.	List the parts of Micro Hydel PV hybrid system?	2	CO4	BTL1

	Part B	M.A	CO	BTL
11.	Explain the need of hybrid energy conversion system with neat sketch.	16	CO5	BTL2
12.	Draw the block diagram of Hybrid renewable energy system with integration wind energy system and solar PV system with neat sketch and explain the operation.	16	CO5	BTL2
13.	Discuss the case study and issues involved in design of Wind PV Hybrid system?	16	CO5	BTL2
14.	Discuss the case study of Micro Hydel PV hybrid system?	16	CO5	BTL2
15.	Discuss the case study of Biomass Diesel-Fuel-cell hybrid system?	16	CO5	BTL2